Exp no - 11b
Date - 12/09/2026
Name - Andryan Durai Jebin

SIMULATION OF RIP (ROUTING INFORMATION PROTOCOL) USING CISCO PACKET TRACER

AIM:

To simulate and configure the Routing Information Protocol (RIP) using Cisco Packet Tracer and study how routers exchange routing information dynamically to establish communication between different networks.

ALGORITHM:

1. Open Cisco Packet Tracer and create a new network topology.
2. Place two routers, two switches and four PCs in the workspace.
3. Connect two PCs to each switch and connect the switches to their respective routers.
4. Connect the two routers using an Ethernet or serial link.
5. Assign IP addresses and subnet masks to all PCs and router interfaces.
6. Configure the default gateway on each PC using the IP address of its router interface.
7. Configure the router interfaces and enable them using the no shutdown command.
8. Enter router configuration mode on each router.
9. Enable RIP using the router rip command.
10. Configure RIP version 2 and advertise all directly connected networks using the network command.
11. Verify the routing tables using the show ip route command.
12. Use the ping command to test communication between PCs in different networks.
13. Observe the exchange of routing information and packet transmission in Simulation mode.
14. Record the results.

IP ADDRESS CONFIGURATION:

Network 1:

PC0 IP Address: 192.168.1.2

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.1.1

Router R1 LAN Interface: 192.168.1.1

Network 2:

PC1 IP Address: 192.168.2.2

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.2.1

Router R2 LAN Interface: 192.168.2.1

Router-to-Router Link:

R1 Interface: 10.0.0.1

R2 Interface: 10.0.0.2

Subnet Mask: 255.255.255.252

ROUTER R1 CONFIGURATION:

enable

configure terminal

router rip

version 2

no auto-summary

network 192.168.1.0

network 10.0.0.0

end

ROUTER R2 CONFIGURATION:

enable

configure terminal

router rip

version 2

no auto-summary

network 192.168.2.0

network 10.0.0.0

end

COMMANDS:

show ip route

show ip protocols

ping 192.168.2.2

OUTPUT:

1. RIP dynamically exchanges routing information between the routers.
2. The routing table displays routes learned through RIP, usually marked with the letter R.
3. The ping command successfully transmits packets between PCs in different networks when the interfaces and RIP configuration are correct.

Sample routing table entry:

R 192.168.2.0/24 [120/1] via 10.0.0.2

Sample ping output:

Reply from 192.168.2.2: bytes=32 time=2ms TTL=126

Reply from 192.168.2.2: bytes=32 time=1ms TTL=126

RESULT:

Thus, RIP was configured and simulated successfully using Cisco Packet Tracer. The routers exchanged routing information dynamically, and communication between different networks was verified using the ping command.